12	(a)	(i)	(X–ray) <u>photon</u> produced when electron/charged particle is stopped/accelerated (suddenly)	B1	
			range of accelerations (in target)	M1	
			hence distribution of wavelengths	A1	[3]
		(ii)	electron gives all its energy to one photon	B1	
			electron stopped in single collision	B1	[2]
		(iii)	de-excitation of (orbital) electrons in target/anode/metal	B1	[1]
	(b)	(i)	aluminium sheet/filter/foil (placed in beam from tube)	B1	[1]
		(ii)	(long wavelength X-rays) do not pass through the body	B1	[1]